ModelIt!

Student Activity Pack

Why Does Your Coffee Get Cold?

The Hidden Math of Thermal Energy Transfer

Grade 06L2 | NGSS MS-PS3-3, MS-PS3-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **If you pour hot water and cold water into the same cup, what temperature do you think it becomes? Why?**

2. **Does a hot drink cool at the same speed the entire time, or does it change? Explain your thinking.**

3. **Why do you think soup cools down faster on a cold winter day than on a warm summer day?**

4. **Draw a graph of what you think happens to the temperature of hot coffee over time.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Hot Object Temperature | Room Temperature | Temperature Difference | Transfer Rate | Object Temperature Over Time

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Hot Coffee in Cold Room

90 degrees C object, 20 degrees C room

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Hot Coffee in Warm Room

90 degrees C object, 35 degrees C room

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Lukewarm Water

40 degrees C object, 20 degrees C room

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Insulation Level — How well the container prevents heat from escaping, slowing the transfer rate

Object Mass — Larger masses store more thermal energy and take longer to change temperature

Material Conductivity — How easily the container material conducts heat, affecting transfer rate

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does hot coffee cool down fast at first but then seem to slow down, and why does it never get colder than the room? Use evidence from your simulation to support your answers.

1. **Why does the cooling curve flatten out instead of continuing straight down?**

2. **Could a hot object ever cool below room temperature without a refrigerator? Why or why not?**

3. **If you double the temperature difference, does the cooling rate double? What does your model show?**

4. **How is the cooling of coffee similar to the way a phone battery drains?**

5. **Why do thermos bottles work so well based on what you learned about transfer rate and temperature difference?**

8 STEM Challenge: Design a Temperature Control System

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YOUR ENGINEERING MISSION

Design a passive cooling or warming system that maintains a beverage within a target temperature range for the longest time possible.

THE SCENARIO

A hospital needs to keep medications at exactly 37 degrees C during transport. Your biomedical engineering team must design a passive thermal system (no electricity) that maintains temperature within 2 degrees of the target for at least 4 hours.

Engineering Design Thinking

1. WHAT INSULATION MATERIALS WILL YOU USE TO SLOW THERMAL ENERGY TRANSFER?

2. HOW WILL YOU PRE-CONDITION YOUR CONTAINER (PRE-HEAT OR PRE-COOL) TO START NEAR THE TARGET?

3. WHAT IS THE OPTIMAL STARTING TEMPERATURE FOR YOUR BEVERAGE TO STAY IN RANGE THE LONGEST?

4. HOW WILL YOU TEST YOUR SYSTEM AND MEASURE ITS PERFORMANCE OVER TIME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Thermal Engineers design heating, cooling, and insulation systems for buildings, vehicles, spacecraft, and medical devices. They apply Newton's Law of Cooling daily. They earn $65,000-$125,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 A cup of hot chocolate at 85 degrees C is placed in a 20 degrees C room. After 10 minutes, the temperature is 50 degrees C. After 20 more minutes, the temperature is 28 degrees C. What explains this pattern?

- **A.** The drink cooled at a constant rate of about 2 degrees per minute
- **B.** The drink cooled faster at first because the large temperature difference drove rapid heat transfer, then slowed as the gap shrank
- **C.** The room temperature increased, slowing the cooling
- **D.** The cup absorbed heat from the room after 10 minutes

2 In the thermal energy model, why can the hot object never cool below room temperature?

- **A.** The container insulates the object and prevents further cooling
- **B.** At room temperature, the temperature difference is zero, so there is no driving force for further heat transfer
- **C.** The object generates its own heat to maintain room temperature
- **D.** Cold air rises away from the object, stopping the cooling process

3 Two identical cups of 90 degrees C water are placed in different rooms. Cup A is in a 10 degrees C room and Cup B is in a 30 degrees C room. After 15 minutes, which statement is true?

- **A.** Both cups lost the same amount of heat because they started at the same temperature
- **B.** Cup A lost more heat because the larger temperature difference drove faster transfer
- **C.** Cup B lost more heat because the warmer room contains more thermal energy
- **D.** Neither cup lost heat because 15 minutes is too short for significant cooling

4 Which graph shape correctly represents how a hot object's temperature changes over time as it cools toward room temperature?

- **A.** A straight line dropping at a constant rate to room temperature
- **B.** A curve that drops steeply at first, then gradually flattens as it approaches room temperature
- **C.** A curve that drops slowly at first, then accelerates as it approaches room temperature
- **D.** A zigzag line that goes up and down randomly

5 A thermos keeps coffee hot for hours while a metal cup lets coffee cool in minutes. Based on the thermal energy model, what does the thermos change?

- **A.** The thermos raises the room temperature around the coffee
- **B.** The thermos eliminates the temperature difference between the coffee and the room
- **C.** The thermos slows the transfer rate by reducing how quickly heat can flow through the container walls
- **D.** The thermos creates new heat energy to replace what is lost